

Chongjie Si

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Shanghai, China

SUMMARY

Chongjie Si is now a Ph.D Candidate at the Artificial Intelligence Institute, Shanghai Jiao Tong University. His research interests lie in efficient LLM pre-training beyond.

EDUCATION

- **Shanghai Jiao Tong University** Sept. 2022 - Present
Ph.D, Artificial Intelligence, supervised by Prof. Wei Shen Shanghai, China
- **Southeast University** Sept. 2018 - Jun. 2022
BSc, Chien-Shiung Wu College, Artificial Intelligence, rank 1 / 121 Nanjing, China

WORK EXPERIENCE

- **Anuttacon** 🌐 Jan 2026 - Now
LLM Research Intern, LLM pre-train, mentored by Hugh Cai Shanghai, China
 - **Optimizer and Training Stability:** Conducted end-to-end validation of Muon across the full LLM pipeline, including pre-training and supervised fine-tuning (SFT); developed optimizer-based solutions to improve training stability under different stages and settings.
 - **Large-Batch Training Stability:** Explored hyper-parameter dynamics under ultra-large batch-size training, and developed stability-oriented solutions to mitigate optimization instability and improve training robustness.
 - **SFT Pipeline for Pre-training:** Built an end-to-end SFT pipeline for pre-training models to evaluate downstream gains introduced by pre-training; performed scaling-law-based hyperparameter search to identify optimal SFT configurations and assess the impact of suboptimal settings.
 - **Math Data Pipeline:** Developed math-oriented data pipelines for both pre-training and mid-training, including organic data cleaning pipelines and synthetic data generation pipelines for large-scale mathematical reasoning corpora.
 - **MoE Load-Balancing Analysis:** Systematically investigated the impact of MoE auxiliary load-balancing objectives on next-token prediction loss; conducted coefficient ablations and evaluated auxiliary-loss-free expert-bias routing strategies to study the trade-off between expert utilization, routing constraints, and benchmark performance.
 - **Dense Shortcut Architecture Validation:** Evaluated block-wise dense shortcut connections for improving deep Transformer trainability, with emphasis on gradient propagation, representation reuse, and pipeline-parallel compatibility; analyzed their effects on training stability and convergence behavior.
 - **Tokenizer Evaluation and Training:** Investigated the impact of existing tokenizers on LLM pre-training, including vocabulary compatibility, special-token alignment, data retokenization, and loss comparison across model variants; further trained customized tokenizers and analyzed how tokenizer-training factors, such as data mixture, corpus scale, regularization, and normalizer design, affect pre-training efficiency and performance.
- **Xiaohongshu** 🌐 Jun 2025 - Dec 2025
REDstar Intern, LLM pre-train, mentored by Dr. Debing Zhang Shanghai, China
 - **Muon Productionization:** Investigated Muon's optimization characteristics through systematic ablations and supported feature integration and optimizer selection; applied Muon to LLM pre-training to improve large-batch stability and reduce compute waste in long training runs.
 - **AdaMuon** 🌐: Designed AdaMuon by combining Muon's Newton-Schulz orthogonalization with adaptive second-momentum scheduling, delivering improved training efficiency and stability. AdaMuon has been integrated into [NVIDIA Emerging-Optimizers Repo](#).
 - **Scaling Law:** Built and maintained scaling-law experiment to support hyper-parameter search and feature-effect validation for pre-training; led the empirical validation of Muon's scaling behavior.
 - **Evaluation Infrastructure:** Assisted infra-side evaluation development, including Megatron-to-HuggingFace checkpoint conversion and launching evaluation jobs with vLLM/SGLang.
- **Alibaba Group** 🌐 Mar. 2025 - Jun. 2025
LLM Research Intern, LLM post-training, mentored by Yadao Wang and Dr. Wenbo Su Hangzhou, China
 - **Efficient SFT:** Investigated efficient fine-tuning strategies for foundation models, aiming to maximize downstream performance gains while updating only a minimal number of trainable parameters.
 - **Model Merging:** Explored model merging techniques to integrate task-specific capabilities into foundation models without additional training or access to original training data.

RESEARCH PROJECTS

- **Efficient Adaptation of Foundation Models** Mar. 2024 – Now
PEFT, LoRA, Subspace Optimization, Model Merging
 - **Subspace-based PEFT:** Studied parameter-efficient adaptation by representing high-dimensional model updates with compact task-relevant subspaces; explored initialization, alignment, direction–magnitude decomposition, and adaptive parameter allocation.
 - **Model Merging:** Investigated how to merge task-specific (LoRA) adapters into a unified model without additional training, focusing on shared subspace construction and pretrained-compatible merge coordinates.
- **Learning with Inaccurate Supervision** Jan. 2022 – Now
Partial Label Learning, Multi-Label Learning, Weak Supervision
 - **Partial Label Learning:** Studied label disambiguation under ambiguous candidate labels and analyzed why accurate models can emerge from inaccurate supervision via spectral and subspace perspectives.
 - **Multi Label Learning:** Investigated robust multi-label learning under incomplete or noisy annotations, with emphasis on high-order label dependencies and sparse high-rank structures.
- **Open-Environment Visual Perception** Jan. 2023 – Now
Open-Vocabulary Segmentation, Weak Supervision, Multimodal Perception
 - **Open-Vocabulary Segmentation:** Explored multimodal and guidance-based methods for semantic segmentation under open category spaces.
 - **Weakly Supervised Segmentation:** Studied segmentation with limited supervision and evolving categories, focusing on adaptation and generalization in open environments.

HONORS AND AWARDS

- **Optiver AI PHD Scholarship (Only 6 Award Winners)** Apr. 2026
Optiver
- **BODE Scholarship (Top 1%)** Oct. 2025
Shanghai Jiao Tong University
- **Doctoral National Scholarship (Top 1%)** Oct. 2024
Chinese Ministry of Education
- **Outstanding Graduate Student** Jun. 2022
Southeast University
- **National Scholarship (Top 1%)** Oct. 2019
Chinese Ministry of Education

ACADEMIC SERVICE

- **Journal Reviewer**
TPAMI, TMI, TCSVT, TKDE
- **Conference Reviewer**
ICML, ICLR, CVPR, ICCV, ECCV, NeurIPS, AAAI, KDD, ARR Rolling...

PUBLICATIONS

Book

- 2025, Wei Shen, **Chongjie Si**, Chen Yang, Yong Yu. *Hands-On Computer Vision*. Posts and Telecom Press & Springer Nature. [🌐 1.2k star]

Journal or Conference

- 2026, ECCV. **Chongjie Si***, Yidan Cui*, Fuchao Yang, Wei Shen. **Why Can Accurate Models Be Learned from Inaccurate Annotations?**
- 2026, CVPR. **Chongjie Si***, Yidan Cui*, Fuchao Yang, Wei Shen. **Revisiting Sparsity Constraint Under High-Rank Property in Partial Multi-Label Learning.**
- 2026, TPAMI. **Chongjie Si**, Zhiyi Shi, Shifan Zhang, Xiaokang Yang, Hanspeter Pfister, Wei Shen. **Task-Specific Directions: Definition, Exploration, and Utilization in Parameter Efficient Fine-Tuning.**
- 2026, ICLR. Muqing Liu*, **Chongjie Si***, Yuheng Jia. **FlexLoRA: Entropy-Guided Flexible Low-Rank Adaptation.**
- 2025. **Chongjie Si**, Debing Zhang, Wei Shen. **AdaMuon: Adaptive Muon Optimizer.**
- 2025. **Chongjie Si**, Xuankun Yang, Muqing Liu, Yadao Wang, Xiaokang Yang, Wenbo Su, Bo Zheng, Wei Shen. **Weight Spectra Induced Efficient Model Adaptation.**
- 2025, NeurIPS, **Spotlight**. Jingjing Jiang, **Chongjie Si**, Jun Luo, Hanwang Zhang, Chao Ma. **Co-Reinforcement Learning for Unified Multimodal Understanding and Generation.**
- 2025, NeurIPS. Xuehui Wang*, **Chongjie Si***, Xue Yang, Yuzhi Zhao, Wenhai Wang, Xiaokang Yang, Wei Shen. **OPMapper: Enhancing Open-Vocabulary Semantic Segmentation with Multi-Guidance Information.**
- 2025, COLM. Zhiyi Shi, Binjie Wang, **Chongjie Si**, Yichen Wu, Junsik Kim, Hanspeter Pfister. **DualEdit: Dual Editing for Knowledge Updating in Vision-Language Models.**
- 2025. **Chongjie Si**, Jingjing jiang, Xiaokang Yang, Wei Shen. **Unveiling the Mystery of Weight in Large Foundation Models: Gaussian Distribution Never Fades.**
- 2025. **Chongjie Si**, Kangtao Lv, Jingjing Jiang, Yadao Wang, Yongwei Wang, Xiaokang Yang, Wenbo Su, Bo Zheng, Wei Shen. **NAN: A Training-Free Solution to Coefficient Estimation in Model Merging.**
- 2025. **Chongjie Si**, Zhiyi Shi, Yadao Wang, Xiaokang Yang, Susanto Rahardja, Wei Shen. **MAP: Revisiting Weight Decomposition for Low-Rank Adaptation.**
- 2025, ICCV. **Chongjie Si**, Zhiyi Shi, Xuehui Wang, Yichen Xiao, Xiaokang Yang, Wei Shen. **Generalized Tensor-based Parameter-Efficient Fine-Tuning via Lie Group Transformations.**
- 2025, ICLR. **Chongjie Si***, Zhiyi Shi*, Shifan Zhang, Xiaokang Yang, Hanspeter Pfister, Wei Shen. **Unleashing the Power of Task-Specific Directions in Parameter Efficient Fine-tuning.**
- 2025, ICLR. **Chongjie Si***, Xuehui Wang*, Xue Yang, Zhengqin Xu, Qingyun Li, Jifeng Dai, Yu Qiao, Xiaokang Yang, Wei Shen. **Maintaining Structural Integrity in Parameter Spaces for Parameter Efficient Fine-tuning.**
- 2024. **Chongjie Si**, Xiaokang Yang, Wei Shen. **See Further for Parameter Efficient Fine-tuning by Standing on the Shoulders of Decomposition.**
- 2024. **Chongjie Si**, Xuehui Wang, Yan Wang, Xiaokang Yang, Wei Shen. **Appeal: Allow Misabeled Samples the Chance to be Rectified in Partial Label Learning.**
- 2024, ECCV. **Chongjie Si**, Xuehui Wang, Xiaokang Yang, Wei Shen. **Tendency-driven Mutual Exclusivity for Weakly Supervised Incremental Semantic Segmentation**
- 2024, AAAI, **Oral**. **Chongjie Si**, Zekun Jiang, Xuehui Wang, Yan Wang, Xiaokang Yang, Wei Shen. **Partial Label Learning with a Partner.**
- 2023, TKDE. **Chongjie Si**, Yuheng Jia, Ran Wang, Min-Ling Zhang, Yanghe Feng, Chongxiao Qu. **Multi-label Classification with High-rank and High-order Label Correlations.**
- 2023, KDD, **Oral**. **Chongjie Si***, Yuheng Jia*, Min-ling Zhang. **Complementary Classifier Induced Partial Label Learning.**